iOS 8th INTERNATIONAL Olympiad of Science 2016-17

CLASS 6

LEVEL - 2

Maximum Marks : 100 | Duration : 40 Minutes | Total Questions: 50 | TEST PAPER TYPE : A D

ENROLLMENT NO: ☐☐☐☐☐☐☐☐☐☐☐☐

CANDIDATE'S NAME : _____

INSTRUCTIONS FOR THE CANDIDATES

1. Write your 12 digit iOS Enrollment Number and your name on top of the question Paper in the given space.
2. Filling up improper Enrollment Number may lead to unavailability of Result.
3. This Question Paper cum Answer Sheet must be submitted to your teacher after the completion of the test.
4. Tick ☑ your answer (A, B, C, D or E) on the given Box ☐ in this Question Paper cum Answer Sheet with HB Pencil or Black/Blue Ball point Pen.
5. The Question Paper Cum Answer sheet Booklet consists of 50 questions, divided into three sections.
6. Section-A: Science (30 questions); Section-B: Mental Ability and Reasoning (10 questions); Section –C: Scholar's Zone (10 questions).
7. Each question of Section-A carries 1.5 marks; Section-B, 2.5 marks each; and that of Scholar's Zone, 3 marks each.
8. Use of calculator or any other devices in the examination is strictly prohibited.
9. This question paper contains 8 pages.
10. Rough work should be done on the blank sheet/space provided in the question paper booklet.
11. Do not start attempting the test paper until you are asked to do so.
12. All the questions are compulsory and there is no negative marking.

SIGN. OF THE TEACHER	SIGN. OF THE STUDENT

DO NOT OPEN THIS QUESTION PAPER BOOKLET UNTIL ASKED TO DO SO

1. Small particles of stone are separated from grains by the method of:
   (A) ☐ Sieving
   (B) ☐ Hand-picking
   (C) ☐ Sedimentation
   (D) ☐ Decantation
   (E) ☐ None of these

2. Cutting a metal stick into small pieces is which one of the following kind of changes?
   (A) ☐ Physical change
   (B) ☐ Chemical change
   (C) ☐ Biochemical change
   (D) ☐ All of these
   (E) ☐ None of these

3. Which one of the following plants stores food in its root?
   (A) ☐ Beet plant (B) ☐ Onion plant
   (C) ☐ Ginger plant (D) ☐ Turmeric plant
   (E) ☐ None of these

4. Which one of the following animals is a scavenger?
   (A) ☐  (B) ☐ 
   (C) ☐  (D) ☐ 
   (E) ☐ None of these

5. Which one of the following is the male part of a flower?
   (A) ☐ Sepal (B) ☐ Petal
   (C) ☐ Stamen (D) ☐ Pistil
   (E) ☐ None of these

6. Which one of the following minerals helps the thyroid gland to make the hormone that controls our growth?
   (A) ☐ Calcium (B) ☐ Phosphorus
   (C) ☐ Iron (D) ☐ Iodine
   (E) ☐ None of these

7. What the thread like structure of a climber plant is called, which coils around any neighbouring object and helps the plant to climbs up?
   (A) ☐ Root (B) ☐ Lamina
   (C) ☐ Tendril (D) ☐ Vascular system
   (E) ☐ None of these

8. In an electric bulb, light is produced due to the glowing of
   (A) ☐ the glass case of the bulb
   (B) ☐ the thin filament
   (C) ☐ the thick wires supporting the filament
   (D) ☐ gases inside glass case of the bulb
   (E) ☐ None of these

9. The pinhole camera works on the principle that:
   (A) ☐ Light travels in a straight line
   (B) ☐ Light travels in a circular line
   (C) ☐ Light travels in a curved line
   (D) ☐ All of these
   (E) ☐ None of these

10. The image formed by a plane mirror is:
   (A) ☐ Real and erect
   (B) ☐ Virtual and erect
   (C) ☐ Real and inverted
   (D) ☐ Virtual and inverted
   (E) ☐ None of these

11. Which one of the following devices is used to open or close a circuit?
   (A) ☐ Switch
   (B) ☐ Fuse
   (C) ☐ Filament
   (D) ☐ Wire
   (E) ☐ None of these

12. In which one of the following methods of disposing of garbage redworms are used?
   (A) ☐ Composting (B) ☐ Vermicomposting
   (C) ☐ Recycling (D) ☐ Reusing
   (E) ☐ None of these

13. Which one of the following fibres is obtained from plants?

(A) ☐ Wool (B) ☐ Silk
(C) ☐ Jute (D) ☐ Nylon
(E) ☐ None of these

14. Which one of the following options is correct if X represents materials soluble in water and Y represents materials insoluble in water?

(A) ☐ X: milk, chalk powder, sugar, common salt; Y: flour, sand, washing soda, alum
(B) ☐ X: alum, copper sulphate, sugar, sand; Y: flour, washing soda, plastic, wax
(C) ☐ X: wax, sugar, alcohol, alum; Y: milk, sand, glass, plastic
(D) ☐ X: milk, sugar, common salt, alcohol; Y: sand, wax, glass, plastic
(E) ☐ None of these

15. Which one of the following options is correct if X, Y and Z represents transparent materials, translucent materials and opaque materials respectively?

(A) ☐ X: air, water; Y: butter paper, ground glass; Z: cardboard, wood
(B) ☐ X: muddy water, ground-glass; Y: air, metals; Z: brick, stone
(C) ☐ X: alcohol, groundnut oil; Y: ground-glass, butter paper; Z: air, brick
(D) ☐ X: plastic, air; Y: muddy water, water; Z: book, brick
(E) ☐ None of these

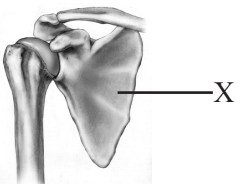
16. Which one of the following processes is responsible for drying up wet clothes when we spread it on a rope in the sunlight?

(A) ☐ Evaporation (B) ☐ Transpiration
(C) ☐ Condensation (D) ☐ Freezing
(E) ☐ None of these

17. When 2 to 3 drops of iodine solution is poured on a freshly cut surface of potato, it gives a blue-black colour. It shows:

(A) ☐ Potato contains starch
(B) ☐ Potato contains proteins
(C) ☐ Potato contains water
(D) ☐ Potato contains vitamins
(E) ☐ None of these

18. Name the part of the shoulder indicated by X in the following figure.



(A) ☐ Acromion (B) ☐ Tuberosity
(C) ☐ Clavicle (D) ☐ Scapula
(E) ☐ None of these

19. Which one of the following statements is correct?

(A) ☐ Yarns are made of fibres
(B) ☐ Fibres are made of yarns
(C) ☐ Yarns are made of fabrics
(D) ☐ Fibres are made of fabrics
(E) ☐ None of these

20. X is one out of the following plants. It has soft, delicate, green coloured stem and its life-span is less than one year. Identify X.

(A) ☐ Rose plant (B) ☐ Lemon plant
(C) ☐ Wheat plant (D) ☐ Hibiscus plant
(E) ☐ None of these

21. What does camel store in its hump?

(A) ☐ Water (B) ☐ Fat
(C) ☐ Ammonia (D) ☐ Sugar
(E) ☐ None of these

22. How many terminals do an electric cell has?

(A) ☐ One (B) ☐ Two
(C) ☐ Three (D) ☐ Four
(E) ☐ None of these

23. The thigh bones of our legs are joined to the hip bone by:

(A) ☐ Hinge joint
(B) ☐ Fixed joint
(C) ☐ Ball and socket joint
(D) ☐ Pivot joint
(E) ☐ None of these

24. Cotton fibres are separated from cotton seeds. This process is known as:

(A) ☐ Ginning (B) ☐ Spinning
(C) ☐ Weaving (D) ☐ Knitting
(E) ☐ None of these

25. Which of the following is correct for respiration in plants?

(A) ☐ Respiration takes place only during day time

(B) ☐ Respiration takes place only during night

(C) ☐ Respiration takes place both during day and night

(D) ☐ Respiration takes place only when plants are not making food

(E) ☐ None of these

26. Which one of the following is an insulator?

(A) ☐ Iron (B) ☐ Gold

(C) ☐ Silver (D) ☐ Paper

(E) ☐ None of these

27. Which of the following statements is incorrect?

(A) ☐ The object having policed surfaces reflects more light than the object having dull surface

(B) ☐ A narrow beam of light is called a ray

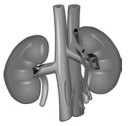
(C) ☐ The light ray which falls on the surface of a object is called reflected ray

(D) ☐ Both (B) and (C)

(E) ☐ None of these

28. Bile, secreted by liver is stored in which of the following organs?

(A) ☐  (B) ☐ 

(C) ☐  (D) ☐ 

(E) ☐ None of these

29. Which one of the following parts of flower contains female sex cells?

(A) ☐ Sepal

(B) ☐ Petal

(C) ☐ Anther

(D) ☐ Ovary

(E) ☐ None of these

30. Name the disease which occurs due to deficiency of vitamin A.

(A) ☐ Marasmus (B) ☐ Rickets

(C) ☐ Night blindness (D) ☐ Scurvy

(E) ☐ None of these

Section–B (Mental Ability and Reasoning)

31. A man and a woman are sitting in a room. Man's mother-in-law and woman's mother-in-law are mother and daughter respectively. Man is the_____ of the woman.

(A) ☐ Father (B) ☐ Father-in-law

(C) ☐ Uncle (D) ☐ Husband

(E) ☐ None of these

32. Ramesh is facing North-East while Suresh is facing exactly the opposite direction what Ramesh is facing. Find in which direction is Suresh facing?

(A) ☐ East

(B) ☐ South

(C) ☐ North-West

(D) ☐ South-West

(E) ☐ None of these

33. In a certain code language 'go and play' is written as 'da pa ta' and 'come and do' is written as 'sa na pa'. Therefore, code for 'and' will definitely be

(A) ☐ pa (B) ☐ da

(C) ☐ ta (D) ☐ sa

(E) ☐ None of these

34. Choose the correct option to fill in the gap.

A __ B B C __ A A B __ C C A __ B B C C

(A) ☐ ACBA (B) ☐ ABBA

(C) ☐ ABCA (D) ☐ ACCA

(E) ☐ None of these

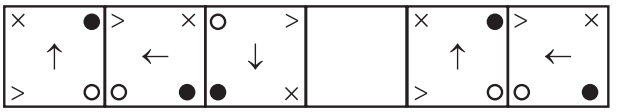
35. If the two hands in a clock are 5 minutes division apart, then the angle between them is:

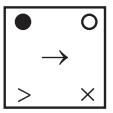
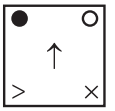
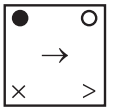
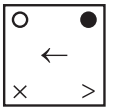
(A) ☐ 3° (B) ☐ 30°

(C) ☐ 24° (D) ☐ 60°

(E) ☐ None of these

36. Fill in the blank box.



(A) ☐ 
(B) ☐ 
(C) ☐ 
(D) ☐ 
(E) ☐ None of these

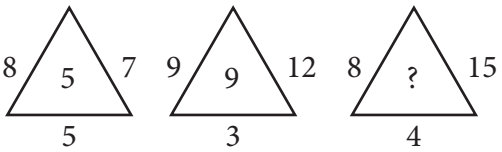
37. If '–' stands for '+', '+' stands for '×', '×' stands for '÷', and ÷ stands for '–', then $10 \times 4 + 5 \div 8 - 3$ is _____.

(A) ☐ 20 (B) ☐ 25
(C) ☐ 23 (D) ☐ 48
(E) ☐ None of these

38. Arrange the following words in Alphabetical order.
Taxi, Technique, Teacher, Technical, Talent

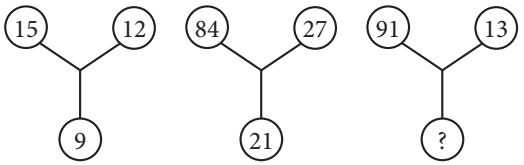
(A) ☐ Talent, Teacher, Taxi, Technique, Technical
(B) ☐ Technique, Technical, Teacher, Taxi, Talent
(C) ☐ Talent, Taxi, Teacher, Technical, Technique
(D) ☐ Talent, Taxi, Teacher, Technique, Technical
(E) ☐ None of these

39. Find the missing number.



(A) ☐ 32
(B) ☐ 28
(C) ☐ 40
(D) ☐ 60
(E) ☐ None of these

40. Find the missing number.



(A) ☐ 14 (B) ☐ 27
(C) ☐ 102 (D) ☐ 99
(E) ☐ None of these

Space for Rough Work

41. Fill in the blanks with correct combination of words.

The presence of cartilage ___p___ between ___q___ makes the ___r___ flexible so that it can bend.

(A) ☐ p-balls, ribs, r-backbone
(B) ☐ p-discs, q-vertebrae, r-backbone
(C) ☐ p-ribs, q-vertebrae, r-backbone
(D) ☐ p-discs, q-vertebrae, r-rib cage
(E) ☐ None of these

42. Fill in the blanks with correct combination of words.

The ___p___ of earth around the ___q___ is an example of ___r___ motion.

(A) ☐ p-rotation, q-moon, r-periodic
(B) ☐ p-revolution, q- jupiter, r-rotational
(C) ☐ p-rotation, q-axis, r-rectilinear
(D) ☐ p-revolution, q- sun, r-periodic
(E) ☐ None of these

43. Which of the following statements is correct?

i. The solid particles which are insoluble in a liquid can be separated by decantation.
ii. A mixture of silver filings and sulphur powder can be separated by using a magnet.
iii. A mixture contains molecules of two or more substances.
iv. Soluble solids can be separated from liquid by evaporation.

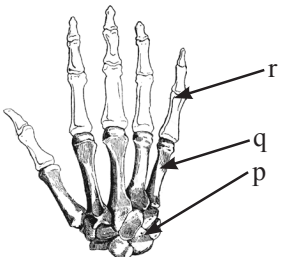
(A) ☐ only (i)
(B) ☐ only (i) and (iii)
(C) ☐ only (i), (iii) and (iv)
(D) ☐ all (i), (ii), (iii) and (iv)
(E) ☐ None of these

44. Match the following:

a.	desert habitat	i.	very cold
b.	grassland habitat	ii.	very little rain
c.	rainforest habitat	iii.	plenty of rain
d.	polar habitat	iv.	moderate rain

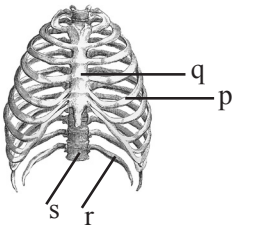
(A) ☐ a→ii, b→iv, c→iii, d→i
(B) ☐ a→ii, b→i, c→iii, d→iv
(C) ☐ a→ii, b→iv, c→i, d→iii
(D) ☐ a→i, b→iv, c→iii, d→ii
(E) ☐ None of these

45. Look at the following figure and choose the correct option.



(A) ☐ p-phalanges, q-carpals, r-metacarpals
(B) ☐ p-metacarpals, q-carpals, r-phallanges
(C) ☐ p-carpals, q-metacarpals, r-phallanges
(D) ☐ p-carpals, q-phallanges, r- metacarpals
(E) ☐ None of these

46. Look at the following figure and choose the correct option.



(A) ☐ p-ribs, q-sternum, r-floating ribs, s-vertebral column
(B) ☐ p-sternum, q-ribs, r-floating ribs, s-vertebral column
(C) ☐ p-vertebral column, q-sternum, r-floating ribs, s-ribs
(D) ☐ p-floating ribs, q-sternum, r-ribs, s-vertebral column
(E) ☐ None of these

47. Which of the following statements is correct?

i. All plants depend on animals for dispersion of their seeds.
ii. The main types of habitats on the earth are aquatic and terrestrial.
iii. The producers and consumers are biotic components of the environment.
iv. Many rainforest animals use camouflage to escape from enemies.

(A) ☐ ii, iii, and iv (B) ☐ i, ii, and iv
(C) ☐ i, iii and iv (D) ☐ i, ii, iii and iv
(E) ☐ None of these

48. Which of the following statements is correct?

(i) During a solar eclipse, the shadow of the moon falls on the earth .

(ii) A pinhole camera forms an inverted image of an object on a screen.

(iii) A shadow is formed when the path of light is obstructed by an opaque object.

(iv) Light travels in a straight line.

(A) ☐ (i) and (iv) only

(B) ☐ (ii) and (iv) only

(C) ☐ (i), (ii) and (iii) only

(D) ☐ all (i), (ii), (iii) and (iv)

(E) ☐ None of these

49. Match the following:

a.	Human	i.	Fins
b.	Cow	ii.	Wings
c.	Snake	iii.	Legs
d.	Eagle	iv.	Whole body
e.	Fish	v.	Limbs

(A) ☐ a→iii, b→iv, c→i, d→v, e→ii

(B) ☐ a→iii, b→v, c→iv, d→ii, e→i

(C) ☐ a→iii, b→i, c→iv, d→ii, e→v

(D) ☐ a→i, b→v,c→iv, d→ii, e→iii

(E) ☐ None of these

50. Which of the following statements is incorrect?

(i) All living things are made up of cells.

(ii) A group of similar living animals that cannot reproduce among themselves are called extinct species.

(iii) The size of an organism is primarily dependent on the number of cells in the body.

(iv) If a potted plant kept near a window from where light is coming in, the tip of the plant grows towards the window.

(A) ☐ (i) and (ii)

(B) ☐ (ii) and (iii)

(C) ☐ only (ii)

(D) ☐ (i), iii) and (iv)

(E) ☐ None of these

Space for Rough Work

ANSWERS

INTERNATIONAL OLYMPIAD OF SCIENCE 2016 (LEVEL 2)

CLASS 6

1.	(B)	2.	(A)	3.	(A)	4.	(D)	5.	(C)	6.	(D)	7.	(C)	8.	(B)	9.	(A)	10.	(B)
11.	(A)	12.	(B)	13.	(C)	14.	(D)	15.	(A)	16.	(A)	17.	(A)	18.	(D)	19.	(A)	20.	(C)
21.	(B)	22.	(B)	23.	(C)	24.	(A)	25.	(C)	26.	(D)	27.	(C)	28.	(E)	29.	(D)	30.	(C)
31.	(B)	32.	(D)	33.	(A)	34.	(A)	35.	(B)	36.	(C)	37.	(E)	38.	(C)	39.	(B)	40.	(A)
41.	(B)	42.	(D)	43.	(C)	44.	(A)	45.	(C)	46.	(A)	47.	(A)	48.	(D)	49.	(B)	50.	(C)